

Atomic Coordinate Lattice Isomorphisms Are of Chain at Most 3

fa_banach_001 packet for arXiv:2507.14297

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Status

This is a partial result for Question 3.4 of Szczepanski's paper [1]. The source proves that every lattice isomorphism on $\ell_p = \ell_p(\mathbb{N})$, $1 \leq p < \infty$, is of chain 3, and then asks whether the same conclusion holds for every Banach lattice. The result below proves the same conclusion for the standard atomic coordinate lattices over an arbitrary infinite index set.

Source Question

The relevant source passage is on page 9 of the locally compiled source PDF.

Recall that an invertible operator $T \in L(\ell_p)$ is a *lattice isomorphism* if for every $x \in \ell_p$ we have $|Tx| = T|x|$, where $|x|$ is the point-wise modulus of x . It is known that every lattice isomorphism on ℓ_p for $1 \leq p < \infty$ is of the form $Te_n = w_n e_{\sigma(n)}$ for some sequence of positive weights $(w_n)_{n=1}^\infty$ and a bijective $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ (see for example Proposition 3.3 in [GDGGT22]). Combining this with Theorem 3.2 gives:

Corollary 3.3. *Let $X = \ell_p$ with $1 \leq p < \infty$ and let $T \in L(X)$ be a lattice isomorphism on X . Then T is of chain 3.*

A natural question is if we can generalise the above Corollary to any lattice isomorphism.

Question 3.4. *Let X be a Banach lattice and $T : X \rightarrow X$ be a lattice isomorphism. Is T of chain 3?*

Transcription: “Let X be a Banach lattice and $T : X \rightarrow X$ be a lattice isomorphism. Is T of chain 3?”

Definitions

Following [1], an operator $T \in L(X)$ is of chain $N \geq 2$ if there are non-scalar operators $S_1, \dots, S_{N-1}$ and a non-zero compact operator K such that

$$T \leftrightarrow S_1 \leftrightarrow S_2 \leftrightarrow \dots \leftrightarrow S_{N-1} \leftrightarrow K,$$

where $A \leftrightarrow B$ means $AB = BA$. Chain 0 means compact, and chain 1 means commuting with a non-zero compact operator.

Proof Intuition

The source paper's countable ℓ_p proof finds a proper subset of basis vectors invariant under the square of the permutation induced by the lattice isomorphism. The same parity idea is purely set-theoretic: every permutation of an infinite set has a nonempty proper subset invariant under its square. For $\ell_p(\Gamma)$ and $c_0(\Gamma)$, lattice isomorphisms must permute coordinate atoms with positive weights, and every coordinate subset gives a bounded band projection. The square of the lattice isomorphism therefore commutes with a non-scalar coordinate projection, and that projection commutes with a coordinate rank-one operator.

Main Result

Theorem 1. *Let Γ be an infinite set and let*

$$X \in \{\ell_p(\Gamma) : 1 \leq p < \infty\} \cup \{c_0(\Gamma)\}.$$

If $T : X \rightarrow X$ is a lattice isomorphism, then T is of chain at most 3. More precisely, either T is connected to a rank-one operator by a chain of length 3,

$$T \leftrightarrow T^2 \leftrightarrow P_B \leftrightarrow F,$$

or T^2 is scalar and T is connected to a rank-one operator by a chain of length at most 2.

Proof. For each $\gamma \in \Gamma$, write e_γ for the coordinate atom. A lattice isomorphism maps positive atoms to positive atoms: if $0 \leq v \leq Te_\gamma$, then $0 \leq T^{-1}v \leq e_\gamma$, hence $T^{-1}v$ is a scalar multiple of e_γ , and so v is a scalar multiple of Te_γ . The positive atoms of $\ell_p(\Gamma)$ and $c_0(\Gamma)$ are exactly the positive scalar multiples of the e_γ 's. Thus there are positive weights w_γ and a bijection $\sigma : \Gamma \rightarrow \Gamma$ such that

$$Te_\gamma = w_\gamma e_{\sigma(\gamma)} \quad (\gamma \in \Gamma).$$

We first record the set-theoretic substitute for the countable lemma in [1]. Fix $\gamma_0 \in \Gamma$ and let $O = \{\sigma^n(\gamma_0) : n \in \mathbb{Z}\}$ be its σ -orbit. If $O \neq \Gamma$, then O itself is a nonempty proper subset invariant under σ^2 . If $O = \Gamma$, then the orbit is infinite, so

$$B = \{\sigma^{2n}(\gamma_0) : n \in \mathbb{Z}\}$$

is nonempty, proper, and invariant under σ^2 . Hence in all cases there is a nonempty proper $B \subset \Gamma$ with

$$\gamma \in B \iff \sigma^2(\gamma) \in B.$$

Let $P_B : X \rightarrow X$ be the coordinate projection $P_B x = 1_B x$. This is a bounded non-scalar projection on both $\ell_p(\Gamma)$ and $c_0(\Gamma)$. On each coordinate atom,

$$T^2 e_\gamma = w_\gamma w_{\sigma(\gamma)} e_{\sigma^2(\gamma)}.$$

The displayed equivalence for B gives $T^2 P_B e_\gamma = P_B T^2 e_\gamma$ for every γ . By density of finitely supported vectors, $T^2 P_B = P_B T^2$ on X .

Choose $\beta \in B$ and define the rank-one coordinate projection

$$F(x) = x_\beta e_\beta.$$

Then F is non-zero compact and $P_B F = F P_B = F$. If T^2 is non-scalar, we therefore have the chain

$$T \leftrightarrow T^2 \leftrightarrow P_B \leftrightarrow F,$$

with both intermediate operators non-scalar.

It remains only to handle the degenerate case where $T^2 = \lambda I$. The formula for $T^2 e_\gamma$ then forces $\sigma^2 = \text{id}_\Gamma$. Since Γ is infinite, the permutation σ has more than one orbit. Let C be a nonempty proper union of σ -orbits. The coordinate projection P_C commutes with T , and a rank-one coordinate projection F_C supported inside C commutes with P_C . Thus

$$T \leftrightarrow P_C \leftrightarrow F_C.$$

This is a chain of length at most 2, and hence the asserted at-most-3 conclusion follows. If one insists on displaying a chain with exactly two non-scalar intermediates, repeat the same projection: $T \leftrightarrow P_C \leftrightarrow P_C \leftrightarrow F_C$. $\square$

Verification Notes

The proof uses only three elementary facts, all checked explicitly above: lattice isomorphisms preserve atoms, every permutation of an infinite set has a proper σ^2 -invariant subset, and coordinate projections on $\ell_p(\Gamma)$ and $c_0(\Gamma)$ are bounded. No numerical computation is used.

Cheap run indexes were searched for ‘2507.14297’, ‘chain of commuting operators’, ‘lattice isomorphism’, ‘chain 3’, and ‘minimal chain’; no duplicate packet or prior attempt for this paper was found. Exact web phrase searches on 2026-07-01 for the arXiv id, title, and the lattice-isomorphism question did not locate a later answer. Novelty confidence is moderate: the proof is a direct cardinality extension of the source paper’s coordinate-projection mechanism, not a solution of the full Banach-lattice question.

Limitations

This packet does not address non-atomic Banach lattices, nor atomic lattices without the full family of bounded coordinate band projections. In particular, it does not settle possible band-ergodic weighted-composition phenomena on function lattices. The remaining hard case is where no non-trivial band projection is visibly invariant under T^2 and no rank-one eigenvector/ eigenfunctional argument is available.

Human Review Recommendation

Check the atom-preservation reduction and the degenerate case $T^2 = \lambda I$. If those are accepted, the result is a clean partial answer extending the source paper’s $\ell_p(\mathbb{N})$ corollary to arbitrary-cardinality standard atomic coordinate lattices.

References

- [1] Tomasz Szczepanski, *On the chain of commuting operators on Banach spaces and Lomonosov’s invariant subspace theorem*, arXiv:2507.14297, 2025.